Temporal Rate Matrix V3.0: A Tested-Effective Weak-Field Transport and Synchronization Framework

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Abstract

This paper summarizes the V3.0 review baseline of the Temporal Rate Matrix / Temporal Quantum Matrix (TRM/TQM) framework as a weak-field effective transport and synchronization model [4, 5]. The framework is organized into scalar, vector, and nonlocal theta sectors, plus a unified effective-action roadmap constrained by guard tests [8]. Across the current test stack, the model is presented as tested-effective and hypothesis-supported, with explicit claim boundaries and open first-principles gaps [6, 7]. This work does not claim theorem-level first-principles closure and does not claim equivalence with General Relativity.

1 Introduction

TRM/TQM V3.0 is positioned as a review-ready scientific baseline: numerically test-gated, falsifiable, and explicitly bounded in scope [4]. The central question is whether the framework is mathematically coherent and reproducible as a weak-field effective model while first-principles closure remains incomplete.

The review focus is therefore not replacement claims, but internal consistency, cross-sector compatibility, and falsifiability under regression and holdout-style guard design [7].

2 Framework Architecture

The architecture consists of four connected layers [5]:

1. Scalar transport backbone (photon transport and EL/Fermat bridge behavior).
2. Vector rotational candidate sector (weak-field frame-dragging structure).
3. Nonlocal theta observable sector $\Theta \to O_5 \to \lambda_\Theta \to g_{\mathrm{obs}}$.
4. Unified effective-action layer linking scalar, vector, and theta sectors under limit-preservation guards.

This architecture is used as a test-gated decomposition, not as a claim of completed microscopic derivation.

3 Scalar, Vector, and Theta Sectors

3.1 Scalar sector

Core object. Effective transport index and memory-channel structure, including the $\phi^2|\dot{\mu}|$ branch [4].

Current physical role. Weak-field transport backbone with executable EL/Fermat bridge behavior and memory-channel consistency.

Test guard block. EL bridge guards, MEM/TRM/MC baseline block, and MC09-MC12 hardening [7, 4].

Claim boundary. Tested-effective/hypothesis-supported in weak-field domain; not theorem-level microscopic closure.

The memory-path sequence used in current interpretation is:

$$A_{\text{dyn}}(\phi) \propto \phi$$

$$I_{\text{micro}} = A_{\text{dyn}}^2|\dot{\mu}|$$

$$I_{\text{micro}} \propto \phi^2|\dot{\mu}|$$

3.2 Vector sector

Core object. Rotational field pair $(\vec{A}_T, \vec{B}_T = \nabla \times \vec{A}_T)$ with effective coupling k_T [5].

Current physical role. Weak-field frame-dragging candidate sector with structural Lense-Thirring-like scaling tests [2].

Test guard block. FD01-FD20, including derived-k_T stability, weak-field compatibility windows, and systematic-bias audits [4, 7].

Claim boundary. Candidate weak-field sector; not full GR-equivalent closure and not theorem-level first-principles derivation.

3.3 Theta sector

Core object. Nonlocal observable chain $\Theta \to O_5 \to \lambda_\Theta \to g_{\text{obs}}$.

Current physical role. Tested-effective candidate path for nonlocal observable response in the current framework [5].

Test guard block. TO01-TO28, TQK01-TQK04, LC01-LC08, and TOL01-TOL04 [4, 7].

Claim boundary. Strongly supported derivation chain at guard level; not theorem-level fundamental closure to g_{obs}.

4 Test-Gated Methodology

4.1 Evaluation protocol

The methodology is test-first and guard-based, with sector-specific hardening followed by cross-sector integration checks:

- Gap 1 hardening: MC09-MC12.
- Gap 2 hardening: RBF21-RBF23.
- Gap 3 hardening: TO/TQK/LC/TOL blocks.
- Gap 4 hardening: FD01-FD20.

- Unified-action integration: UF01-UF09.

Category gates are used operationally as fast hard gate (`CoreRegression`), default validation (`Category!=LongRunning`), and extended sweeps (`Category=LongRunning`).

4.2 Acceptance criteria

At this stage, acceptance is defined by:

1. sector guard pass under fixed documented criteria;
2. limit-preservation behavior under ablations/holdouts where applicable;
3. no cross-sector contradiction in unified guard slices;
4. explicit retention of claim boundaries in interpretation text.

4.3 Regression/guard philosophy

The guard philosophy is conservative: preserve previously validated behavior, add falsifiable new blocks incrementally, and treat unification/cross-terms as candidate-level unless stability and limit guards remain satisfied [8].

5 First-Principles Gaps

The first-principles status remains open in documented form [6]:

- Gap 1 (memory term): strongly supported path, not theorem-level microscopically closed.
- Gap 2 ($m = 3$ closure): strongly constrained/hardened path, not theorem-level microscopic proof.
- Gap 3 (theta observable chain): strongly supported as tested-effective, not theorem-level closure.
- Gap 4 (vector k_T normalization): strongly hardened weak-field path, not theorem-level first-principles closure.

6 Unified Action Roadmap

The unified-action path is documented as a candidate-level roadmap [8]:

$$S_{\text{eff}}[T, \vec{A}_T, \Theta] = S_T[T] + S_A[\vec{A}_T] + S_\Theta[\Theta] + S_{\text{int}}[T, \vec{A}_T, \Theta].$$

UF01-UF09 currently support:

- scalar/vector/theta limit recovery,
- zero-coupling additive decomposition,
- bounded small cross-couplings,
- global cross-coupling identifiability without per-group refit dependency,
- preservation of MC/FD/TO guard behavior under small couplings.

This is a candidate-level unified effective-action roadmap, not theorem-level unification.

7 Claim Boundaries

The V3.0 claim boundary is explicit and enforced:

- TRM/TQM is presented as tested-effective and hypothesis-supported in weak-field scope [4].
- It does **not** claim to replace General Relativity [3].
- It does **not** claim theorem-level first-principles completion [6].
- It does **not** claim full GR-equivalent closure across all sectors [2].

8 Conclusion

TRM/TQM V3.0 provides a consolidated review baseline with numerical hardening, explicit cross-sector structure, and clear non-overclaim boundaries.

The next falsification steps are direct:

1. failure of unified cross-sector guards (UF block) under new external benchmark regimes;
2. failure of microscopic derivation paths (memory, $m = 3$, theta chain) to remain structurally consistent under tighter constraints;
3. failure of weak-field benchmark compatibility in independent external comparisons (including frame-dragging/scalar transport references) [2, 1].

Any of these failures would require narrowing, revising, or rejecting parts of the current effective-framework interpretation.

References

[1] Federico Lelli, Stacy S. McGaugh, and James M. Schombert. Sparc: Mass models for 175 disk galaxies with spitzer photometry and accurate rotation curves. *The Astronomical Journal*, 152(6):157, 2016.

[2] Josef Lense and Hans Thirring. On the influence of the proper rotation of central bodies on the motions of planets and moons according to einstein's theory of gravitation. *Physikalische Zeitschrift*, 19:156–163, 1918. Historical frame-dragging reference; verify final bibliographic formatting.

[3] Placeholder. General relativity benchmark tests and comparisons (placeholder), 2026. Replace with concrete external GR benchmark reference(s) before submission.

[4] TRM Project. Trm current status for peer review, 2026. Repository document: docs/review/TRM_Current_Status_For_PeerReview.md.

[5] TRM Project. Trm field sector map, 2026. Repository document: docs/Theory/TRM_Field_Sector_Map.md.

[6] TRM Project. Trm first-principles gap list, 2026. Repository document: docs/Theory/TRM_First_Principles_Gap_List.md.

[7] TRM Project. Trm real physics test coverage, 2026. Repository document: docs/review/TRM_Real_Physics_Test_Coverage.md.

[8] TRM Project. Trm unified field action roadmap, 2026. Repository document: docs/Theory/TRM_Unified_Field_Action_Roadmap.md.